

RiskPulse

An AI/NLP risk engine that turns news and social posts into structured risk signals, then acts on them: a sentiment-tilted index and event-triggered stress tests.

203,363

real news & social
documents, 2021–22

A + B

both downstream
modules built

1

command to run,
CPU only, no keys

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REPLAYED IN TIME ORDER • FEBRUARY 2022

**RiskPulse flagged the escalation
two days before the full
invasion.**



22 Feb • 00:00 UTC

Russia recognises separatist regions
Impact 8 → stress test fires



22 Feb • 06:00 UTC

Troop order
Impact 9 → stress test fires again



24 Feb

Full-scale invasion begins

Real 2021–22 news and tweets, replayed with no look-ahead; synthetic USD 10 bn book. One episode, not proof of skill: across the year, trigger timing matched random days (slide 7).

A headline becomes a risk signal only once it has an entity, an event type and a severity

THE PROBLEM



Speed

Markets reprice within hours, but risk teams still read news in batches.



Structure

Raw text has no ticker, no event type and no severity, so nothing downstream can act on it.



Trust

Credit and risk decisions need signals that are explainable, auditable and tested against baselines.

THE APPROACH

AI/NLP Risk Engine • three fields per signal

Sentiment

-1 to +1,
scored per company

Event class

10 types, incl. the
brief's five

Impact

1-10, learned from
abnormal returns

e.g. "Exxon posts record quarterly profit while Chevron cuts its outlook"
→ **XOM +0.64**, **CVX -0.94** (a whole-sentence score of -0.95 would misread Exxon)



Module A • Tactical

subscribes to sentiment

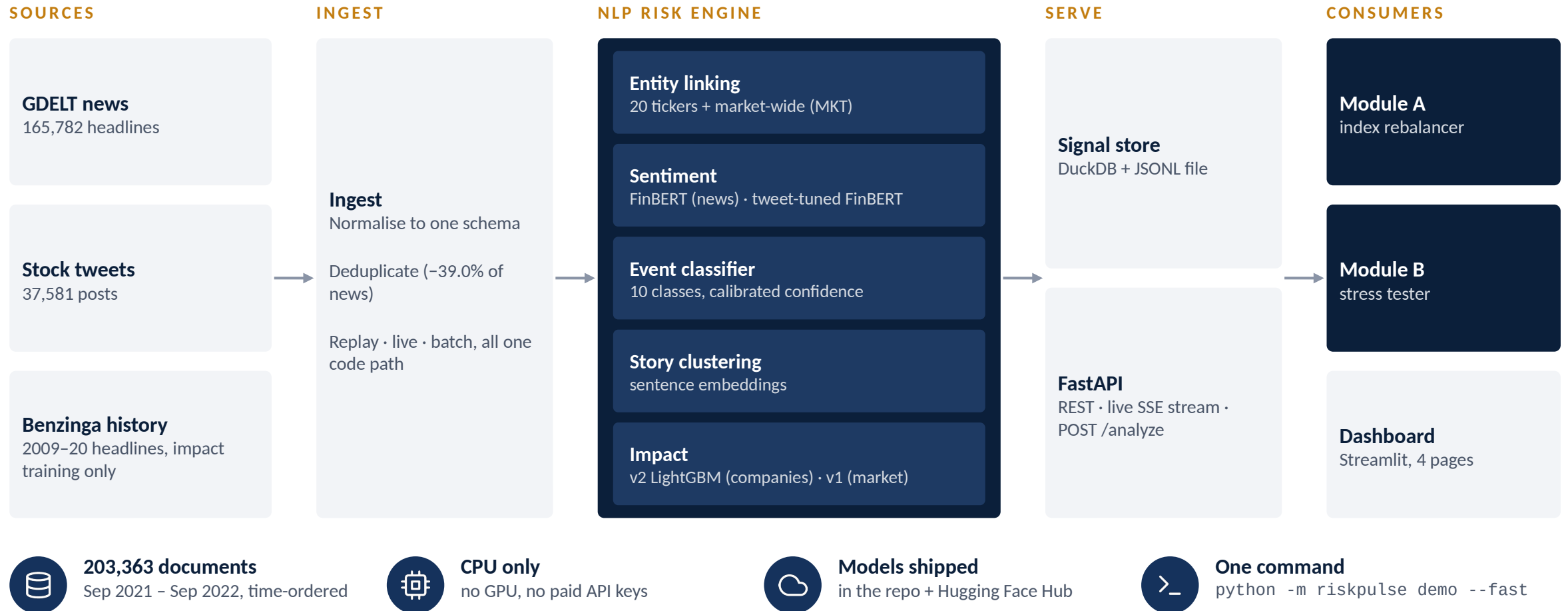
Sentiment-tilted index of 20 S&P 100 stocks,
rebalanced as news arrives.

Module B • Strategic

subscribes to event class + impact

Adverse, high-impact events trigger stress
tests on a USD 10 bn synthetic wholesale
book.

One engine, two consumers: a reproducible pipeline from raw text to stress test on a laptop CPU



Four design choices that make the signals defensible in front of a risk committee



Entity-level sentiment, routed by source

What Scores only the sentences that mention each company. FinBERT handles news; a tweet-fine-tuned FinBERT handles tweets.

Why “Exxon posts record quarterly profit while Chevron cuts its outlook” scores XOM +0.64 and CVX -0.94; the whole sentence (-0.95) misreads Exxon. Routing chosen on 397 labelled news items.



Impact learned from market reactions

What Monotone LightGBM on the seven formula drivers (event prior, sentiment, velocity, breadth, novelty, relevance, credibility) plus mention count, trained on 2009–18 headlines vs 2-day abnormal returns.

Why “Impact” has no ground-truth label, so the market supplies one. Every score shows its drivers.



A stress engine in a risk team's language

What Transaction-seeded synthetic book of loans, bonds, swaps, FX, options and CDS, repriced with duration/convexity, DV01 and PD $\times$ LGD $\times$ EAD, with CET1 from Basel CRE20 weights.

Why Shocks are calibrated only on pre-September-2021 episodes, so 2022 is a genuine out-of-sample test.



A rebalancer built like a real mandate

What Tilt $w \cdot \exp(\kappa \cdot s \cdot c)$, with κ fixed by a 1.5% active-weight risk budget, never by returns. Daily turnover capped at 5%; trading costs charged.

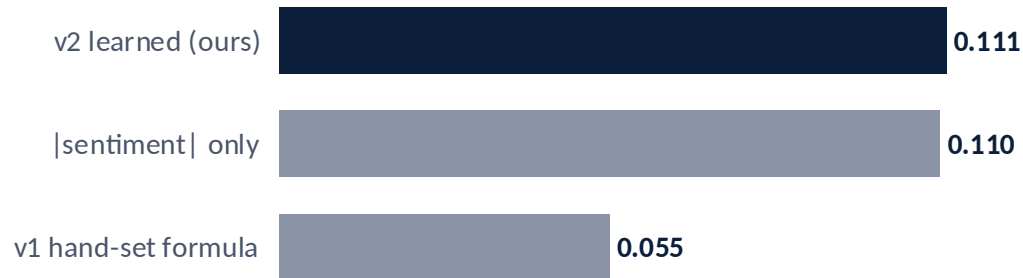
Why Avoids overfitting the test year and keeps turnover realistic for a real index.

Discipline: decision rules pre-registered before each test · every look at the 2021–22 test window logged · 500 human labels + 100 entity checks

Stack: Python 3.11 · transformers (FinBERT) · sentence-transformers · LightGBM · DuckDB · FastAPI + SSE · Streamlit + Plotly · pytest

Measured against naive baselines on human-labelled and out-of-sample data; every look at the test window is logged

Impact vs realised |abnormal return|, Spearman ρ (2021–22 test, 3,538 ticker-days)



Sentiment on human-labelled live news, macro-F1 (n = 237)



9 / 14

market-factor directions right after the 2022 invasion trigger

Out-of-sample: shocks calibrated on pre-Sep-2021 episodes only

0.047

information coefficient, sentiment vs next-day return (t = 2.85, 209 days)

Tilt strength fixed without looking at returns

82.0%

entity-linking precision (95% CI 73.3–88.3%)

100 human-checked links

86.6/s

documents per second; 21.3 ms p95 per document

Laptop CPU, no GPU

Gains vs naive: learned impact $\approx 2.0\times$ the hand-set score's link to market moves • FinBERT +0.121 macro-F1 over VADER • dedup removes 39.0% of news

Built for the people who act on risk: explainable stress tests with a full audit trail

INSIDE THE 24 FEB 2022 STRESS RUN

–USD 10.7 m

predicted P&L on the
USD 10 bn synthetic book

13.00% → 12.98%

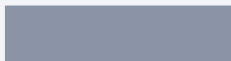
CET1 ratio,
before → after

Every shock is traced to a named historical analogue: **Crimea 2014 • Brexit 2016 • Abqaiq 2019**

Predicted vs realised, next 10 sessions

9 of 14 factor directions right. The biggest miss: oil.

Predicted



Oil +6.5%

Realised



Oil +18.0%

WHO ACTS ON IT



Credit and risk teams **Module B**

An adverse, multi-source event triggers a stress test with P&L, Δ EL and CET1, each shock traced to its analogue.



Portfolio managers **Module A**

Sentiment-tilted index weights with the headlines behind every move. Turnover capped at 5% a day; cost drag 0.5%.



Analysts **Signal Monitor**

One screen shows why a story scores 8/10: drivers, sources and evidence. Any headline can be scored live.

All counterparties are synthetic (CP_XXXX). No client or confidential data is used. Source: reports/metrics.json

Where it breaks, why, and what comes next

What we found	Likely cause	Next step
Event classification is level with a keyword baseline (two rounds, 500 hand labels, one blind test)	Noisy headlines spread across 10 classes	Active-learning labels on live news
Market-wide impact does not predict SPY or VIX moves ($\rho \approx 0$)	The learned model covers companies only; market events use the formula	Train a market-level impact model on index moves
Oil underestimated after the invasion: +6.5% predicted vs +18.0% realised	Analogues mix risk-off with commodity-supply shocks	Separate supply-shock and risk-off scenarios
June 2022 FOMC hike missed	Scheduled events are priced in; the market rallied in relief	Surprise-vs-consensus measure for scheduled releases
Trigger timing no better than random: precision 26.4% vs 26.6% for random days; fires on 71% of sessions	The threshold is relative (top 5% of items); with this much news it is crossed on most days	A fixed alert budget, pre-registered; a market-level impact model

HOW WE KEPT IT HONEST

- 500 human labels and 100 entity checks, labelled by the author
- Decision rules pre-registered before each test
- Every look at the 2021–22 test window logged
- Weak results reported, not tuned away
- Every design decision logged in DECISIONS.md

“We used 2022 to find where the method breaks, not to tune it until it looks right.”